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The complete machine-learning workflow

Week 3: scikit-learn end to end, and the four ways a pipeline silently lies to you.

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By the end of this week you should be able to

Learning outcomes served: LO1, LO3, LO5

- Construct a complete scikit-learn Pipeline from raw data to a saved model.
- Identify data leakage in a described protocol, and name which kind it is.
- Choose an appropriate cross-validation scheme, including when grouping matters.
- Establish and report a baseline, and interpret a model relative to it rather than to zero.
- Make an analysis reproducible: seeds, versions, and a fixed protocol declared before looking at results.

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The biological problem

Part 1 · Why anyone cares about this week's method

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Biological context

- A biological ML result is only as trustworthy as the protocol that produced it. The failure modes in this week are not exotic: they are the ordinary mistakes that produce impressive published numbers which do not reproduce.

- A review of machine-learning-based science found 329 papers across 17 fields affected by data leakage. In biology specifically, selecting features on the whole dataset before cross-validation has been known to produce dramatically optimistic error estimates since at least 2002.

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The method

Part 2 · Concepts, then the mathematics

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The workflow

Problem definition, data collection, exploration, preprocessing, train/test split, training, hyperparameter tuning, evaluation, interpretation, deployment. The order is not decorative: preprocessing decisions made after seeing the test set are leakage.

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Data leakage

Information from outside the training data reaching the model. It inflates the reported score without improving the model, so it is invisible unless you look for it.

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Key concepts

Preprocessing leakage Fitting a scaler, imputer or feature selector on the full dataset before splitting. The fix is to put every fitted step inside a Pipeline.

Duplicate and group leakage The same patient, sample or near-duplicate row appearing in both training and test. Use GroupKFold on the patient identifier, not the row index.

Temporal leakage Training on data collected after the prediction time. If the model will be used prospectively, evaluate it prospectively.

Cross-validation Stratified k-fold for classification; grouped variants when rows are not independent; nested CV when you are also tuning hyperparameters.

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Key concepts (continued)

Baselines The majority class on WDBC gives 62.7 per cent accuracy. A model reporting 85 per cent has beaten nothing worth beating. Always report the baseline alongside.

Reproducibility Fixed random seeds, recorded library versions, a script rather than a notebook run out of order, and the protocol written down before the results are seen.

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k-fold cross-validation

$$CV(k) = (1/k) \cdot \Sigma \; L(M_{-i} , S_i)$$

Partition the data into k folds. Fit on k−1 of them, evaluate on the one held out, and average. Stratified folds preserve the class balance in every fold.

- M_{-i} the model fitted with fold i held out
- S_i fold i, used only for evaluation
- L the chosen loss or score
- k 5 or 10 in practice; larger k means less bias, more variance and more compute

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What leakage actually costs, measured

Every figure produced by data/scripts/02_build_benchmarks.py. Note rows two and four: the inflation is sometimes zero.

Protocol	Reported	Honest	Inflation
Feature selection on 5,000 noise features, random labels	88.0%	55.0%	33 points
Feature selection on the real WDBC data	94.2%	94.2%	0 points
25% of rows duplicated before splitting, 1-NN	97.2%	95.1%	2.1 points
Scaler fitted on all rows before splitting	97.4%	97.4%	0 points

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The taxonomy of leakage

Kapoor and Narayanan (2023) found 329 papers across 17 fields affected by leakage.

- Preprocessing leakage: a scaler, imputer or selector fitted before the split
- Feature-selection leakage: columns chosen using all the labels
- Group leakage: several rows from one patient, site or batch split across folds
- Temporal leakage: training on data collected after the prediction time
- Target leakage: a feature that is a consequence of the outcome, such as a medication prescribed because of the diagnosis
- Duplicate leakage: the same or near-identical row on both sides
- Selection on the test set: tuning a hyperparameter against the data you then report

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The rules that prevent it

You cannot detect leakage from the score. Only from the protocol.

- Split first. Everything that learns anything from data goes inside a Pipeline
- Use GroupKFold on the patient identifier when rows are not independent
- Establish a baseline before believing any model: the majority class scores 62.7% on WDBC
- Fix seeds, record library versions, and write the protocol down before you see the result
- Touch the test set exactly once

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Worked example

Part 3 · How much does leakage actually cost you?

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How much does leakage actually cost you?

- This is worth measuring rather than asserting, so the numbers below were produced by running the protocols on real data. The script is in the repository at data/scripts/02_build_benchmarks.py.
- Case one: 100 samples, 5,000 features of pure Gaussian noise, and labels assigned at random. There is no signal at all, so an honest protocol must report chance. Selecting the best 20 features on the whole dataset and then cross-validating gives 88 per cent accuracy.
- Selecting inside each fold gives 55 per cent. The first number is entirely manufactured.
- Case two: the real WDBC data, feature selection done the wrong way. The inflation is essentially zero, because the informative features get selected either way. This is the more unsettling result.

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How much does leakage actually cost you? (continued)

- The size of the inflation depends on the data, so the absence of a large gap is not evidence that your protocol was sound. You cannot detect leakage by looking at the score.

- Case three: duplicating a quarter of the rows before splitting, with a 1-nearest-neighbour classifier, raises measured accuracy from 95.1 to 97.2 per cent. The model has learned nothing; it is being tested on copies of its own training data.

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Leakage is not detectable from the score. It is only detectable from the protocol.

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Try this interactively

Train/Test Leakage Simulator

How much does a leaky protocol inflate the reported score?

`weeks/week03.html` · `Interactive exploration`

Open the week page on the course website and scroll to Interactive exploration.

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The foundation-model connection

Part 4 · Leakage gets worse, not better, with foundation models

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Leakage gets worse, not better, with foundation models

- Two new failure modes appear once pretrained models are involved. First, pretraining-set contamination: if the model was pretrained on a corpus that already contains your test sequences, your held-out set was never held out.
- Checking this is genuinely difficult, because pretraining corpora are large and not always fully documented.
- Second, embedding leakage: computing embeddings for the whole dataset is fine, because the foundation model never sees your labels, but any label-aware step applied afterwards, such as selecting embedding dimensions by their correlation with the target, must still go inside the fold.

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Interpretation and practical

Part 5 · What the result means, and what you will do with it

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A colleague reports 0.97 AUC for predicting treatment response from gene expression, using 5-fold cross-validation on 80 patients with 20,000 genes. Before you believe the number, what three questions do you ask about the protocol, and what answer to each would make you stop worrying?

Think like a scientist. Bring your answer to the laboratory.

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Building a pipeline that does not lie to you

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This week's laboratory · Construct a leak-free scikit-learn Pipeline, then deliberately break it and measure the damage. · 2 hours

- Build a Pipeline of StandardScaler, SelectKBest(k=5) and LogisticRegression. Evaluate with stratified 5-fold cross-validation.
- Now do it wrongly on purpose: fit SelectKBest on the full dataset first, then cross-validate the reduced data. Compare the two figures.
- Repeat both protocols on a synthetic dataset of 100 samples, 5,000 standard-normal features and random labels. The true accuracy is 0.5 by construction. Record what each protocol reports.
- Explain why the inflation is dramatic in task 3 and negligible in task 2. What does that tell you about using the score to detect leakage?
- Duplicate 25 per cent of the WDBC rows, then cross-validate a 1-NN classifier with and without the duplicates. Explain the gap.
- Write down the protocol you would pre-register for a real study using this data: the split, the metric, the baseline, the seeds, and what you would do if the result disappointed you.

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Self-check

- You impute missing values using the column mean computed over the whole dataset, then split into train and test. What is wrong?
- A) Nothing, mean imputation is standard
- B) Test-set information has entered the training data through the imputed values
- C) The mean is the wrong statistic
- D) You should have dropped the rows

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Self-check

- Your dataset has three tissue samples per patient. You use standard 5-fold cross-validation on the rows. What is the risk?
- A) Class imbalance
- B) Group leakage: samples from one patient appear in both training and test folds
- C) Too few folds
- D) Overfitting the hyperparameters

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References and further reading

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Questions

Next week: continue to Week 4.

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